

Attack-Type Analysis — VAE and Dense Autoencoder, Canonical Model (19 Features)

Technical report — prepared for Gérard

Scope: `06_attack_type_analysis/` , `11_pairwise_segmented_v2/` ,
`09_dense_v2_comparison/` , `10_vae_v2_comparison/` ,
`14_concurrency_feature_experiment/`

1. Context and Objectives

This report documents the current canonical anomaly-detection pipeline: a VAE and a Dense autoencoder, both trained on **19 modeling features** — the original 18 (packet/byte/duration statistics, protocol/service/conn-state one-hots) plus one new feature, `concurrency_src_1s` , added on 2026-07-30 after an investigation into why one attack type was being missed. There is no v1/v2 split in this report: every number below is the current model. The prior 18-feature results are archived in full at `V1_ARCHIVE/` for anyone who needs the before/after comparison; this document reports the current state directly.

Three evaluation protocols are used throughout, identical for both models:

1. **Single attack-type**: each attack type evaluated against the full benign test pool, the other two types excluded from that run.
2. **Pairwise**: each of the 3 attack-type pairs evaluated against benign, the third type excluded — includes a *decomposed* (per-constituent-type) recall to separate "recall moves because the mix changed" from "recall moves because the model's behavior on that type changed."
3. **Segmented (contiguous-block) injection**: the same test flows, reordered into one stream — benign → apache_bench → benign → slowloris → benign → portscan → benign — to check whether contiguous placement (vs. shuffled) changes per-flow outcomes.

`threshold_95` (95th percentile of reconstruction error on held-out benign-only validation flows, recalibrated per seed) is the operating point for recall/F1/FPR throughout. VAE scoring is **deterministic** ($z = z_{\text{mean}}$, no reparameterization noise — see the Methodology Note).

2. VAE Model Results

Model: `phase3_vae/05_contamination_sweep/04_models/contam_0pct` , 5 seeds, 19 features, deterministic z_{mean} scoring.

2.1 Single attack-type

attack_type	n_attack	ROC-AUC	PR-AUC	F1 (thr95)	Benign FPR (thr95)	Attack Recall (thr95)
apache_bench	1487	0.9836 ± 0.0123	0.9035 ± 0.0805	0.8428 ± 0.0337	0.0664 ± 0.0110	0.9500 ± 0.0453
portscan	694	0.9998 ± 0.0001	0.9983	0.7551	0.0664 ± 0.0110	1.0000 ± 0.0000
slowloris	929	1.0000 ± 0.0000	1.0000	0.8048	0.0664 ± 0.0110	1.0000 ± 0.0000

(source: `10_vae_v2_comparison/results_single_attack_type_vae_v2.csv/.md`)

2.2 Pairwise combinations

Pair	n_attack	ROC-AUC	PR-AUC	F1 (thr95)	Pooled Recall (thr95)
portscan + apache_bench	2181	0.9887 ± 0.0084	0.9641	0.8888	0.9659 ± 0.0309
portscan + slowloris	1623	0.9999 ± 0.0000	0.9997	0.8779	1.0000 ± 0.0000
apache_bench + slowloris	2416	0.9899 ± 0.0076	0.9715	0.8989	0.9692 ± 0.0279

Decomposed apache_bench-only recall (the same per-flow flag decision, isolated to apache_bench flows regardless of what else is in the eval set): solo = 0.9500 ± 0.0453; paired with portscan = 0.9500 ± 0.0453; paired with slowloris = 0.9500 ± 0.0453 — **identical up to seed noise in all three settings**, confirming the pooled-recall increase in the table above is a mixing artifact (adding well-detected portscan/slowloris flows pulls the pooled number up), not an actual change in how apache_bench is detected.

(source: `11_pairwise_segmented_v2/vae/results.md`)

2.3 Segmented (contiguous-block) injection

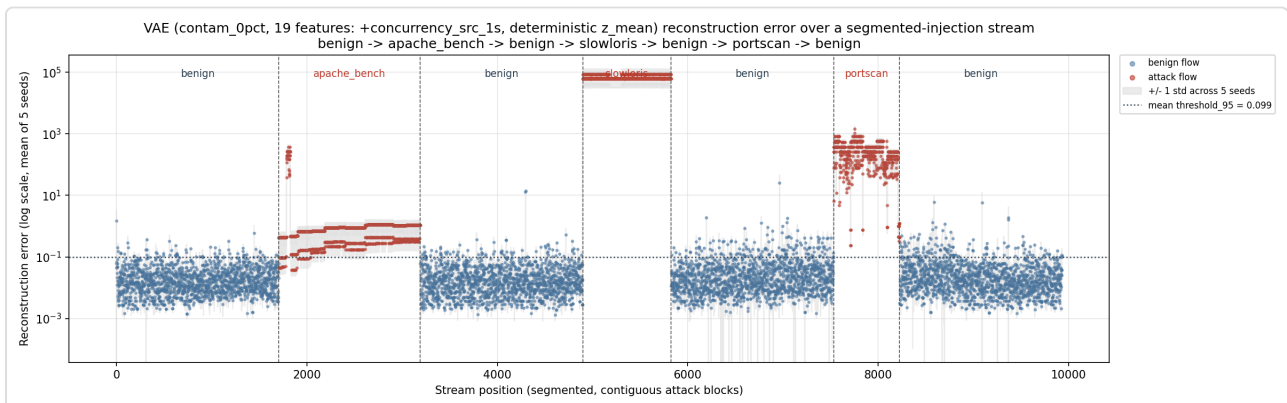


Figure 1 — Reconstruction error (log scale, mean of 5 seeds) vs. stream position. Dashed vertical lines mark segment boundaries; the dotted horizontal line is the mean threshold_95 (0.0987). All three attack blocks now sit visibly above the benign band.

Segment	n	Benign FPR (thr95)	Attack Recall (thr95)	F1 (thr95)	Recall in shuffled test (ref.)
benign (seg. 0)	1705	0.0386 ± 0.0166	—	—	—
apache_bench	1487	—	0.9500 ± 0.0453	0.9739 ± 0.0242	0.9500 ± 0.0453
benign (seg. 2)	1705	0.0490 ± 0.0191	—	—	—
slowloris	929	—	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
benign (seg. 4)	1705	0.0967 ± 0.0097	—	—	—
portscan	694	—	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
benign (seg. 6)	1706	0.0811 ± 0.0117	—	—	—

Block recall matches the shuffled-test-set number exactly for every type — consistent with a static per-flow threshold decision with no sequence memory. Benign FPR ranges 0.0386–0.0967 across the 4 gaps (a modest-n sampling spread around the pooled 0.0664 average, not a drift effect).

(source: `11_pairwise_segmented_v2/vae/block_recall_f1.md`)

3. Dense Model Results

Model: `phase3_dense/04_phase3_models/full_features` , 5 seeds, 19 features.

3.1 Single attack-type

attack_type	n_attack	ROC-AUC	PR-AUC	F1 (thr95)	Benign FPR (thr95)	Attack Recall (thr95)
apache_bench	1487	0.9808 ± 0.0076	0.8930	0.8218	0.0660 ± 0.0036	0.9092 ± 0.0382
portscan	694	0.9997 ± 0.0002	0.9973	0.7551	0.0660 ± 0.0036	1.0000 ± 0.0000
slowloris	929	1.0000 ± 0.0000	1.0000	0.8050	0.0660 ± 0.0036	1.0000 ± 0.0000

(source: `09_dense_v2_comparison/results_single_attack_type_dense_v2.csv/.md`)

3.2 Pairwise combinations

Pair	n_attack	ROC-AUC	PR-AUC	F1 (thr95)	Pooled Recall (thr95)
portscan + apache_bench	2181	0.9868 ± 0.0052	0.9587	0.8747	0.9381 ± 0.0261
portscan + slowloris	1623	0.9999 ± 0.0001	0.9995	0.8782	1.0000 ± 0.0000
apache_bench + slowloris	2416	0.9882 ± 0.0046	0.9672	0.8862	0.9441 ± 0.0235

Decomposed apache_bench-only recall: solo = 0.9092 ± 0.0382 ; paired with portscan = 0.9092 ± 0.0382 ; paired with slowloris = 0.9092 ± 0.0382 — same invariance confirmed on the second architecture.

(source: `11_pairwise_segmented_v2/dense_v2/results.md`)

3.3 Segmented (contiguous-block) injection

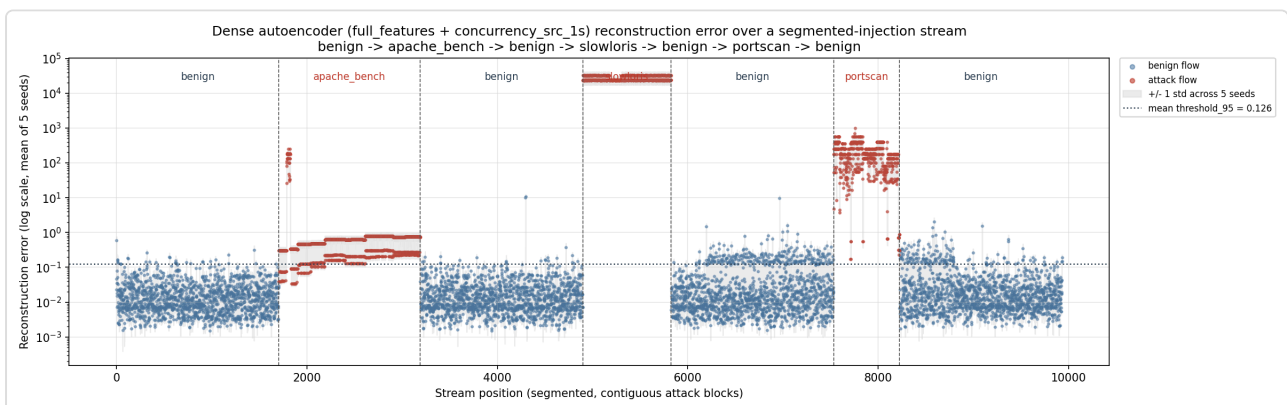


Figure 2 — Same stream as Figure 1, evaluated with the Dense autoencoder. Mean threshold₉₅ = 0.1256.

Segment	n	Benign FPR (thr95)	Attack Recall (thr95)	F1 (thr95)	Recall in shuffled test (ref.)
benign (seg. 0)	1705	0.0151 ± 0.0027	—	—	—
apache_bench	1487	—	0.9092 ± 0.0382	0.9521 ± 0.0211	0.9092 ± 0.0382
benign (seg. 2)	1705	0.0142 ± 0.0061	—	—	—
slowloris	929	—	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
benign (seg. 4)	1705	0.1478 ± 0.0051	—	—	—
portscan	694	—	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
benign (seg. 6)	1706	0.0870 ± 0.0063	—	—	—

(source: `11_pairwise_segmented_v2/dense_v2/block_recall_f1.md`)

3.4 Both architectures agree

apache_bench recall is now 0.91–0.95 depending on architecture (previously 2.6–3.3%), and the effect is consistent across every protocol tested (single, pairwise-decomposed, segmented) and across two architectures trained on genuinely different data (Dense: windows 01–08, 23,274 benign train flows; VAE: window_10 only, 3,049 benign train flows). This consistency is itself evidence that the fix addresses a *feature-space* limitation rather than an architecture- or training-data-specific quirk (see Section 4).

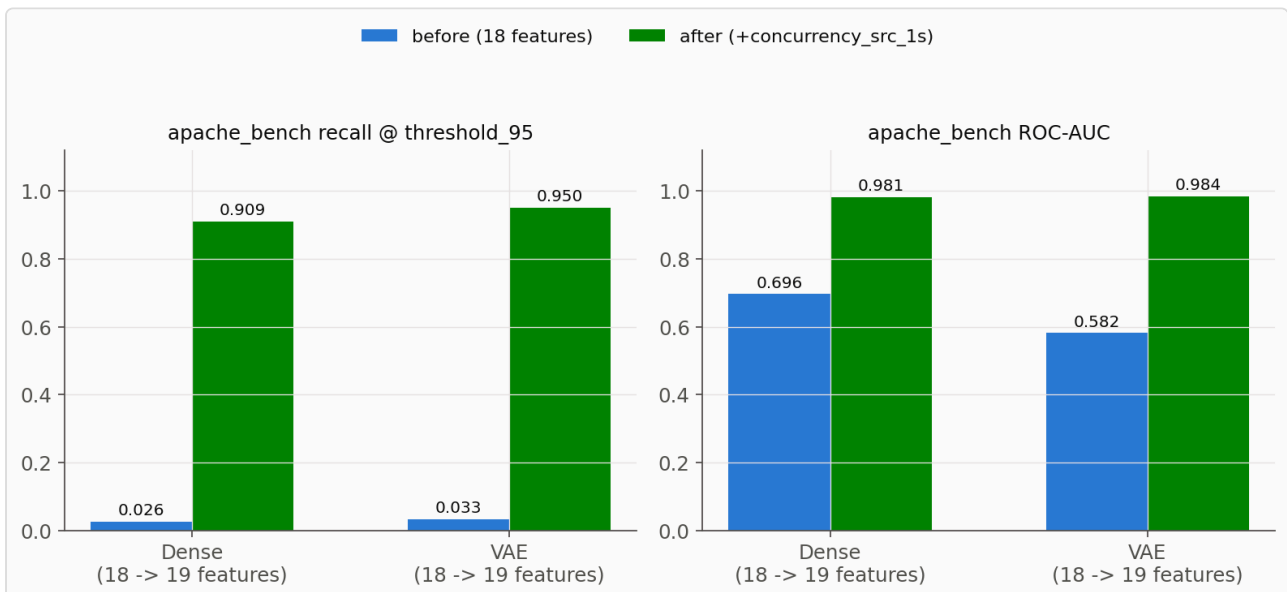


Figure 3 — *apache_bench* recall and ROC-AUC, 18-feature (before) vs. 19-feature (after), both architectures.

4. Root Cause Analysis

4.1 Why `apache_bench` was originally missed

`apache_bench` (`ab`) fires many near-identical short HTTP GET requests. At the single-flow level, each of those requests is an ordinary, unremarkable short HTTP connection — it does not look anomalous on its own. Diagnostic work (Kolmogorov-Smirnov separability + mean-shift analysis on the original 18 features, full detail in `V1_ARCHIVE/10_final_report/04_apache_bench_diagnostics/findings.md`) showed why this defeated reconstruction-error-based detection specifically:

- The best individual features (`orig_pkts_scaled` , `orig_bytes_scaled` , etc.) had **high KS statistics (0.62–0.76)** against benign — on paper, separable.
- But their **mean shift was only 0.4–0.7 benign standard deviations** — i.e., `apache_bench`'s cluster sits *inside* the normal range of benign traffic, not in its tail. The high KS came from `apache_bench` forming a very narrow, low-variance cluster (a stereotyped, repeated request), which produces a sharp empirical-CDF jump even when the cluster's center is unremarkable.
- Reconstruction error sums squared per-feature deviations from a benign-fit manifold. A point sitting inside the training distribution's normal range reconstructs cleanly **regardless of how sharply its CDF differs from benign's** — which is exactly what happened: the typical `apache_bench` flow's reconstruction error was close to the typical benign flow's, and both models flagged only a small, fixed subset of `apache_bench` flows (2.6–3.3% recall).

By contrast, `portscan` and `slowloris` push their best features tens to hundreds of benign standard deviations away (near-categorical splits, e.g. `conn_state=REJ` , extreme `byte_ratio`) — obviously anomalous at the single-flow level, which is why both models already detected them at $\geq 98.8\%$ recall under the 18-feature model.

4.2 What was actually missing, and the fix

What makes `apache_bench` anomalous is not any single request's shape — it's that the **same request repeats far more often, with far less inter-arrival variance, than organic traffic**. That is a property of the *local request rate*, invisible to a model that scores one flow at a time against 18 purely per-flow features.

The fix, `concurrency_src_1s` : for each flow, the count of other flows from the **same source IP** arriving within ± 1 second (log1p-transformed, standardized on the benign-train split only — same leakage-free convention as every other feature). This is purely temporal/volumetric and contains no hardcoded IP value anywhere in its computation. Adding this single 19th feature is what produced the recall jump documented in Sections 2–3.

4.3 Generalization caveat

This lab dataset has exactly one attacker IP and a small number of benign hosts. `concurrency_src_1s` is IP-agnostic by construction (it groups by whatever source IP a flow actually has, not a hardcoded value), but the dataset itself cannot distinguish "high per-source request rate is inherently suspicious" from "high per-source request rate happens to correlate with the one IP that is the attacker here." Knock-out ablation in `14_concurrency_feature_experiment/` (freezing the feature to its benign-train mean at inference time, same trained weights) confirmed the recall gain collapses back to the 18-feature baseline — the model is genuinely using this feature, not something else correlated with it — but that check cannot rule out the dataset-composition caveat above. Before generalizing this result to a deployment with many legitimate high-rate clients (e.g., NAT'd traffic, load balancers, legitimate bursty clients), the feature's benign false-positive behavior should be validated in a multi-source-IP setting.

5. Methodology Note — How the Right Feature Was Found

The path to `concurrency_src_1s` was not direct, and the detour is worth recording briefly. The first hypothesis tested (`13_temporal_feature_experiment/`) was **inter-arrival time relative to the single previous flow** from the same source IP — motivated by an early diagnostic finding that `apache_bench`'s median IAT was $\sim 2364\times$ shorter than benign's. Retraining with that feature **did not move recall at all** (KS = 0.375 on the full dataset, weaker than the existing best features), and exposed that the original $2364\times$ figure was itself a measurement artifact of computing IAT over a sparse test subset rather than the full flow history. The feature that actually worked, tested next (`14_concurrency_feature_experiment/`), was a **local window count** rather than a single-previous-flow gap — the two are related but not equivalent, and only the windowed version captured the signal strongly enough to move detection. Full negative and positive results, including a knock-out ablation and a dataset-composition confound found and corrected along the way, are archived in both experiment folders and summarized in `V1_ARCHIVE/README.md`.

6. Known Limitation

The O3 dedup-prevalence correction has not yet been applied to the current (19-feature) PR-AUC/F1 numbers. The prior 18-feature report applied a correction (recomputing PR-AUC/F1 on a deduplicated test set, since the resampled windows — `window_resampled_15pct/20pct` — repeat real flows and distort prevalence) after confirming ROC-AUC/recall/FPR are behavior metrics unaffected by it (<0.02 difference, see `V1_ARCHIVE/10_final_report/01_single_attack_type/vae/dedup_sanity_check/`). That

correction has not been re-run on the current model. **ROC-AUC, recall, and FPR figures in this report are unaffected**; the PR-AUC/F1 numbers above should be read as provisional until the correction is reapplied — this is flagged as a follow-up item, not a known error.

7. Supplementary Notes (carried forward, still valid)

VAE latent-dimension note. The VAE's bottleneck (8 units) is narrower than its nominal latent dimension (10), so `z_mean` cannot carry more than 8 effective degrees of freedom regardless of the declared latent size. A dedicated ablation (`phase3_vae/05_contamination_sweep/12_latent_ablation/`, run on the 18-feature model) found `latent=8` and `latent=10` statistically indistinguishable on detection metrics — an architectural imprecision without measurable effect on any number reported here.

Threshold calibration note. `threshold_95` is estimated from a small held-out benign validation set (653 flows for the VAE, 4,609 for Dense) — a tail percentile with real sampling noise. Prior analysis (`10_final_report/06_scripts/o4_threshold_transfer/`, run on the 18-feature model) found `threshold_95` varies with $CV \approx 28\%$ across VAE seeds, and the `val`→`test` FPR transfer carries a small systematic upward bias (realized $\approx 5.8\%$ vs. 5.0% nominal). This affects the threshold-dependent operating point (recall/F1/FPR); ROC-AUC/PR-AUC are unaffected.

Threat-model note. Throughout this project, `is_attack` is defined by source-IP identity (`id.orig_h == <attacker IP>`), not by flow behavior. Both models therefore learn "the statistical signature of traffic from the attacking machine," not a behavior-based notion of maliciousness. Scenarios where that equivalence breaks — IP spoofing, NAT'd or shared-source traffic mixing legitimate and malicious flows, lateral movement from a previously trusted host — are not covered by this evaluation and are not tested.

8. Conclusion

1. **The apache_bench detection gap is closed.** Recall rose from 2.6–3.3% to 90.9–95.0% (Dense / VAE respectively), ROC-AUC from 0.58–0.70 to 0.98+, with a small benign-FPR cost (+0.4 to +0.5 percentage points on average). `portscan` and `slowloris` remain at $\geq 99.8\%$ recall, unaffected.
2. **The fix is a feature-space fix, not a model fix.** Two architectures trained on disjoint data (different windows, $\sim 7.6\times$ different training-set size) show the same magnitude of improvement from the same single added feature — the strongest evidence available that the original 18-feature set, not either architecture, was the limiting factor.
3. **The improvement is robust across evaluation protocols.** Decomposed pairwise recall and segmented (contiguous-block) recall both match the shuffled single-attack-type

number exactly, for both models — the detection behavior is a stable per-flow property, not an artifact of how the test set happens to be assembled.

4. **Two things remain open**, tracked explicitly rather than silently assumed resolved: the O3 dedup-prevalence correction has not been reapplied to PR-AUC/F1 (Section 6), and `concurrency_src_1s`'s generalization to multi-attacker or high-legitimate-rate environments is untested on this single-attacker-IP dataset (Section 4.3).